(ADAM) CHEOL WOO KIM

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ACADEMIC APPOINTMENTS

Harvard University — Boston, MA
Postdoctoral Fellow in Computer Science — *Sep 2024 – Present*
• Advisor: Milind Tambe

EDUCATION

Massachusetts Institute of Technology — Cambridge, MA
PhD in Operations Research; GPA: 5.0/5.0 — *Aug 2024*
• Advisor: Dimitris Bertsimas
• Thesis: *Predictive and Prescriptive Trees for Optimization and Control Problems*

Seoul National University School of Law — Seoul, Republic of Korea
J.D. Candidate (withdrew to pursue PhD at MIT) — *Feb 2019 – Jul 2019*

Seoul National University — Seoul, Republic of Korea
B.A. in Economics (Summa Cum Laude); Major GPA: 4.24/4.30 — *Feb 2019*
• Concentration in Mathematical Economics and Microeconomic Theory

RESEARCH INTERESTS

AI Alignment, AI Safety, Human-AI Interaction,
Learning to Optimize, LLM-Assisted Decision Modeling

PAPERS

*: equal contribution; α-β: alphabetical order

Preprints

1. Many Preferences, Few Policies: Compact Portfolios for Multi-objective LLM Alignment
Cheol Woo Kim*, Jai Moondra*, Roozbeh Nahavandi, Andrew Perrault, Milind Tambe, Swati Gupta
Under review at *International Conference on Learning Representations (ICLR) 2027*.
Workshop version: *Conference on Language Modeling (COLM), Workshop on Scientific Understanding of Foundation Models*, 2026.

Journal Papers

1. Optimal Control of Multiclass Fluid Queueing Networks: A Machine Learning Approach
(α-β) Dimitris Bertsimas, **Cheol Woo Kim**
Operations Research (forthcoming), 2026.

2. Generative Artificial Intelligence for Social Impact
Lingkai Kong, **Cheol Woo Kim**, Davin Choo, Milind Tambe
IEEE Intelligent Systems, 2026.

3. Optimal Control of Fluid Restless Multi-armed Bandits: A Machine Learning Approach
(α-β) Dimitris Bertsimas, **Cheol Woo Kim**, José Niño-Mora
Machine Learning, 2026.

4. A Machine Learning Approach to Two-stage Adaptive Robust Optimization
(α-β) Dimitris Bertsimas, **Cheol Woo Kim**
European Journal of Operational Research, 2024.

5. A Prescriptive Machine Learning Approach to Mixed-Integer Convex Optimization
(α-β) Dimitris Bertsimas, **Cheol Woo Kim**
INFORMS Journal on Computing, 2023.

Conference Papers

1. Bilevel Optimization of Synthetic Trajectories for Multi-Turn LLM Fine-Tuning
Shresth Verma, Mauricio Tec, **Cheol Woo Kim**, Kai Wang, Milind Tambe
Advances in Neural Information Processing Systems (NeurIPS), 2026.

2. Incentive-Aware AI Safety via Strategic Resource Allocation: A Stackelberg Security Games Perspective
Cheol Woo Kim*, Davin Choo*, Tzeh Yuan Neoh, Milind Tambe
International Conference on Autonomous Agents and Multiagent Systems (AAMAS), Blue Sky Ideas Track, 2026.

3. Preference Robustness for DPO with Applications to Public Health
Cheol Woo Kim*, Shresth Verma*, Mauricio Tec*, Milind Tambe
AAAI Conference on Artificial Intelligence (AAAI), AI for Social Impact Track, 2026.

4. Navigating the Social Welfare Frontier: Portfolios for Multi-objective Reinforcement Learning
Cheol Woo Kim*, Jai Moondra*, Shresth Verma, Madeleine Pollack, Lingkai Kong, Milind Tambe, Swati Gupta
International Conference on Machine Learning (ICML), 2025.
Journal extension in preparation for submission to *Operations Research*.

5. Robust Optimization with Diffusion Models for Green Security
Lingkai Kong, Haichuan Wang, Yuqi Pan, **Cheol Woo Kim**, Mingxiao Song, Alayna Nguyen, Tonghan Wang, Haifeng Xu, Milind Tambe
Uncertainty in Artificial Intelligence (UAI), 2025.

Workshop Papers

1. Generating Robust Portfolios of Optimization Models using Large Language Models
Eleni Straitouri, **Cheol Woo Kim**, Milind Tambe
International Conference on Machine Learning (ICML), LM4Plan Workshop, 2026.

2. Lightweight Robust Direct Preference Optimization
Cheol Woo Kim*, Shresth Verma*, Mauricio Tec*, Milind Tambe
Conference on Neural Information Processing Systems (NeurIPS), Workshop on Reliable ML from Unreliable Data, 2025.

3. LLM-based Agent Simulation for Maternal Health Interventions: Uncertainty Estimation and Decision-focused Evaluation
Sarah Martinson, Lingkai Kong, **Cheol Woo Kim**, Aparna Taneja, Milind Tambe
International Conference on Autonomous Agents and Multiagent Systems (AAMAS), Workshop on Autonomous Agents for Social Good, 2025.

WORK EXPERIENCE

Microsoft Research Redmond, WA
Research Intern, Machine Learning and Optimization Group *May 2023 – Aug 2023*
• Mentors: Ishai Menache, Marco Molinaro, Konstantina Mellou
• Research: Combinatorial optimization under uncertainty for GPU resource allocation.

Permanent Mission of the Republic of Korea to the United Nations New York, NY
Research Intern, UN General Assembly Group *Jan 2014 – Apr 2014*

TEACHING EXPERIENCE

Developed course materials, led recitations, and held office hours for the following courses:

TA, The Analytics Edge (MIT 15.071) *Spring 2024*
• MBA course on statistics, machine learning, optimization, and data science.

TA, Optimization Methods (MIT 6.7200J/15.093J) *Fall 2022*
• Master's/PhD course on linear, nonlinear, combinatorial, robust, and semi-definite optimization.

TA, Introduction to Mathematical Programming (MIT 6.251J/15.081J) *Fall 2021*
• PhD course on linear, nonlinear, discrete, combinatorial, robust, and semi-definite optimization.

TA, Robust Modeling, Optimization, and Computation (MIT 1.142J/15.094J) *Spring 2021*
• Master's/PhD course on theory and applications of robust optimization.

AWARDS AND HONORS

Kwanjeong Educational Fellowship *Sep 2019 – Feb 2025*
• Merit-based PhD scholarship.

Eminence Scholarship *Aug 2017*
Department of Economics, Seoul National University Seoul, Republic of Korea
• Awarded to top 1% in the economics department.

INVITED TALKS

Workshop on Learning Algorithms for Decision-Making in Games with Mixed Agents, Cornell University *Apr 2026*
• *Towards Scalable Language Model Personalization*

KAIST Graduate School of Data Science Special Seminar *Aug 2024*
• *Machine Learning for Optimization and Control Problems*

INFORMS Annual Meeting *Oct 2023*
• *Optimal Control of Multiclass Fluid Queueing Networks: A Machine Learning Approach*

PROFESSIONAL SERVICE

Journal Reviewer: Journal of Machine Learning Research, INFORMS Journal on Computing, INFORMS Journal on Optimization, European Journal of Operational Research, Transportation Science
Conference Reviewer: AAAI 2026

GRANT PROPOSAL CONTRIBUTIONS

Contributed substantially to writing the following funded proposal:

- **Interactive Strategic Policy Design using LLM Agents**
  Army Research Office; PI: Milind Tambe *Aug 2026 – Jul 2029*

STUDENTS ADVISED

Robert Debbas, Harvard University *2024 – Present*
Shresth Verma, Harvard University *2024 – Present*
Sarah Martinson, Harvard University *2024 – 2025*
Next position: Ph.D. student in Applied Mathematics, Harvard University.

SKILLS

Programming: Python (PyTorch, scikit-learn), Julia, R
Optimization Software: JuMP, Gurobi, MOSEK
Languages: Korean (native), English (fluent)